SDLC Phase 2: System Design
Alumni Management System

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1 Introduction

The Alumni Management System is a web-based platform designed to manage alumni information, communication, and reporting. The system allows administrators to manage alumni records, viewers to generate reports, and alumni to update their personal information.

The system aims to centralize alumni data such as academic history, employment records, and contact details while ensuring secure role-based access. This improves data accessibility, reporting efficiency, and alumni engagement.

2 High-Level System Architecture

The system follows a standard web application architecture consisting of users, a frontend interface, backend services, and a database.

2.1 Architecture Overview

The system includes the following major components:

- Users (Admin, Viewer, Alumni)
- Web Browser Interface
- Frontend Application
- Backend API
- Database System
- External Email/Survey Services

Users access the system through a browser. The frontend communicates with the backend using API requests. The backend processes requests, performs validation, and interacts with the database to retrieve or store information.

2.2 Architecture Diagram

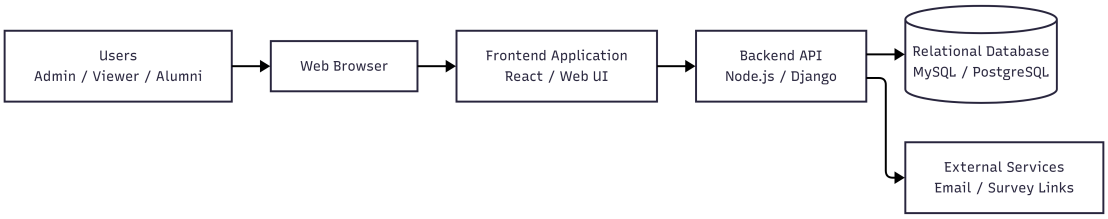

Figure 1: High-Level System Architecture

3 Component Description

3.1 Frontend

The frontend is responsible for presenting the user interface and allowing users to interact with the system.

Responsibilities include:

- Displaying dashboards and alumni profiles
- Providing forms for updating alumni information
- Allowing administrators to manage alumni records
- Allowing viewers to generate reports
- Communicating with the backend via HTTP API requests

The frontend ensures that users have a smooth and intuitive interface for performing system operations.

3.2 Backend

The backend handles the core logic of the system.

Responsibilities include:

- Processing requests from the frontend
- Implementing authentication and role-based access control
- Validating user input
- Managing alumni records and communication campaigns
- Generating reports

The backend exposes APIs that allow the frontend to retrieve and modify system data securely.

3.3 Database

The database stores all persistent system data.

Key stored data includes:

- Alumni profiles
- Degree information
- Employment history

- Internship history
- Organization memberships
- Messages sent to alumni
- User authentication information

A relational database is used to maintain structured relationships between different entities such as alumni, employers, and degree programs.

4 Data Model (Class Diagram)

The system data model represents entities and relationships involved in managing alumni information.

Important entities include:

- User
- Alumni
- Degree
- Employer
- Internship
- Organization
- Message

Each entity contains attributes that store specific information related to alumni records, academic background, employment history, and system communications.

Relationships include:

- One user manages one alumni profile
- Alumni may have multiple degrees
- Alumni may have multiple employment records
- Alumni may belong to multiple organizations
- Admin users can send messages to multiple alumni

4.1 Class Diagram

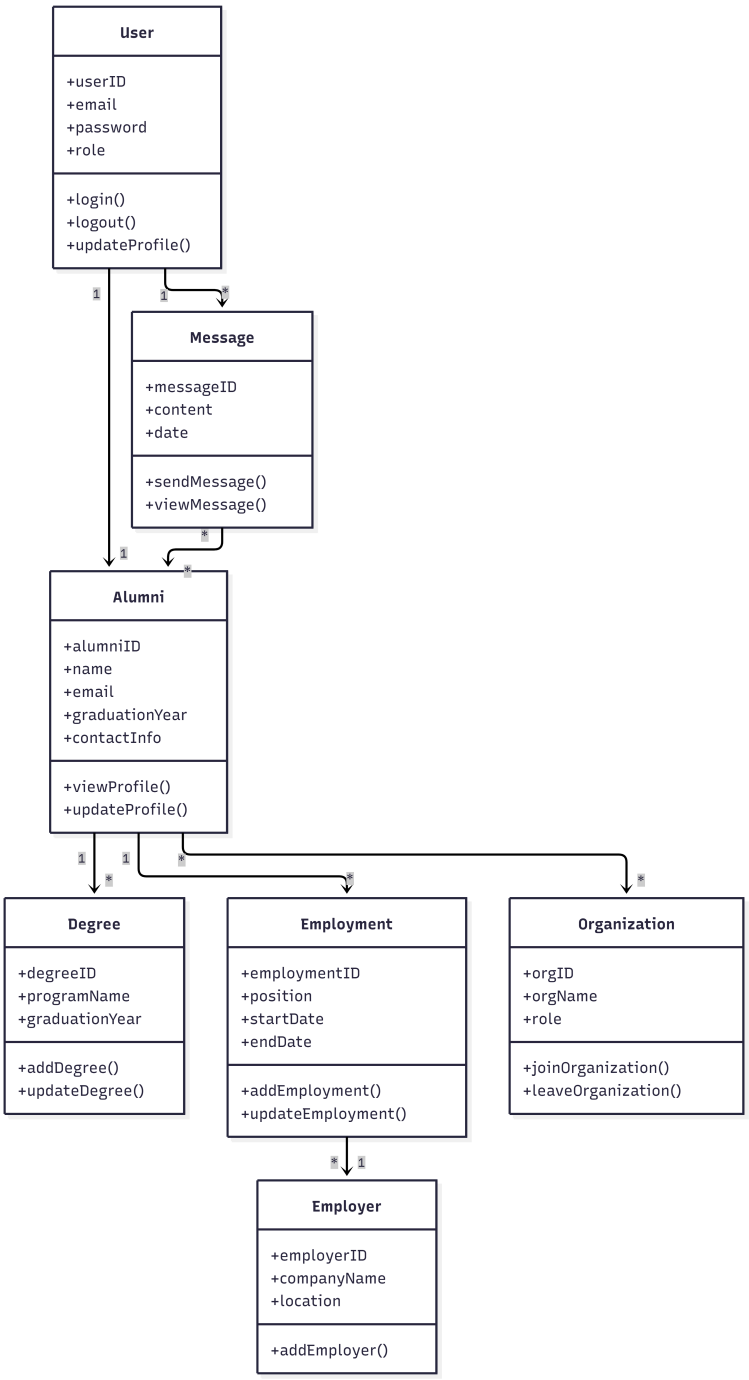

Figure 2: System Class Diagram

5 UI Wireframes

Low-fidelity wireframes represent the layout and structure of the user interface without focusing on design details.

The wireframes help visualize how users will navigate and interact with the system.

5.1 Main Page

The main page includes:

- Login form
- Navigation menu
- System introduction

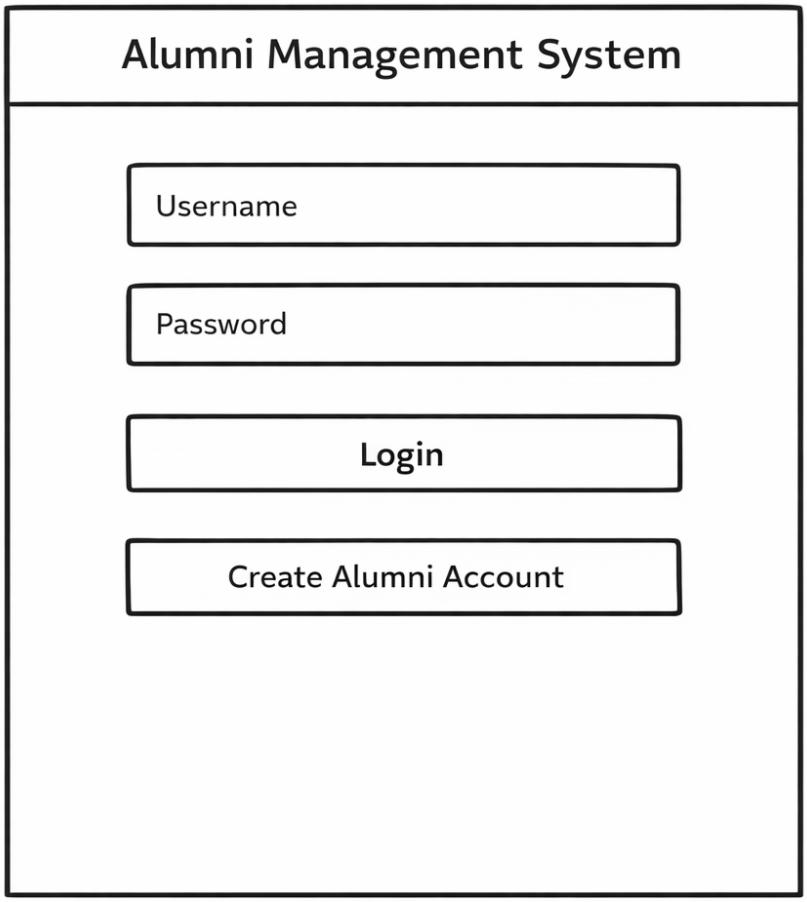

Figure 3: Main Page Wireframe

5.2 Dashboard

The dashboard allows users to view system summaries and access major features.

Features include:

- Navigation panel
- Alumni search
- Reports access
- Profile management

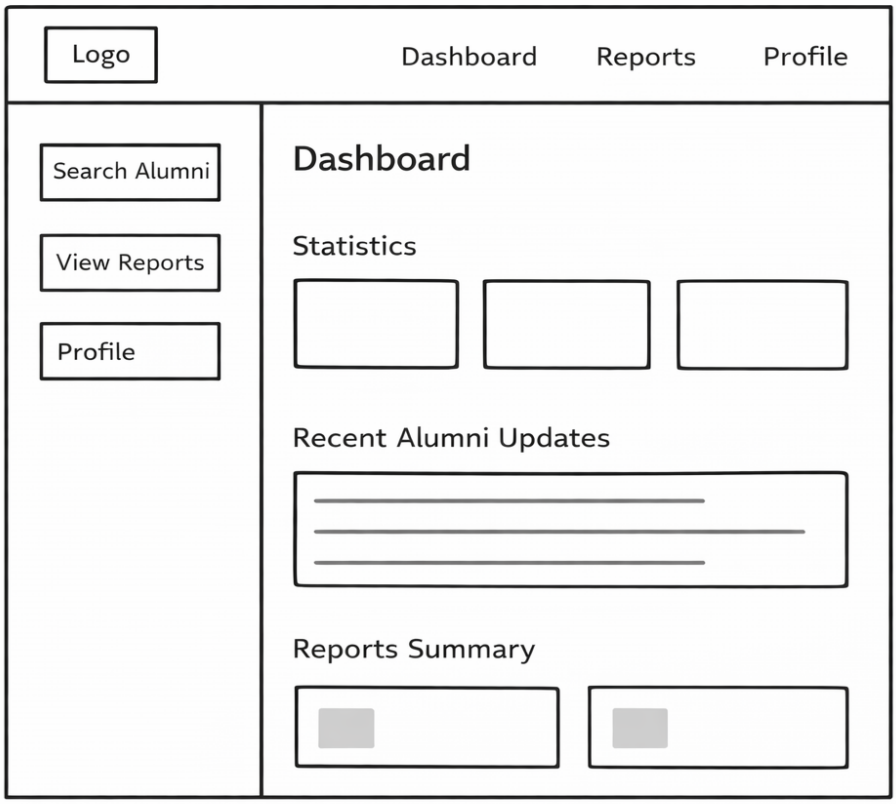

Figure 4: Dashboard Wireframe

5.3 Feature Page (Manage Alumni)

The manage alumni page allows administrators to:

- Add alumni records
- Edit alumni information
- Delete alumni records
- Import alumni data

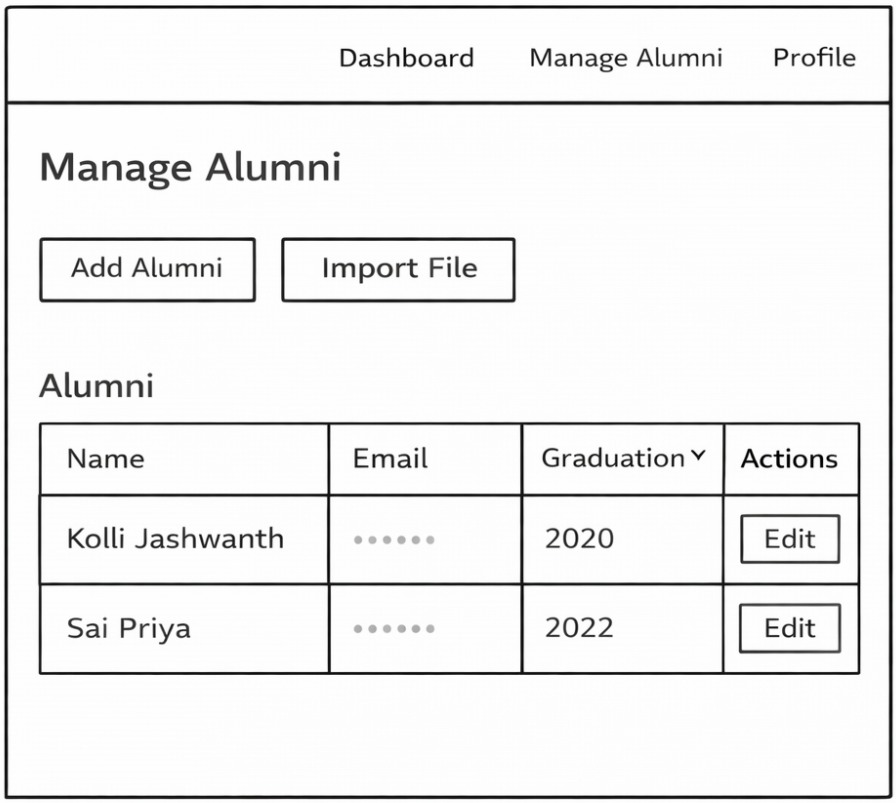

Figure 5: Manage Alumni Wireframe

6 Technology Stack Justification

6.1 Frontend Technology

The frontend can be implemented using modern JavaScript frameworks such as React or Angular. These technologies allow developers to build dynamic and responsive user interfaces that interact efficiently with backend APIs.

6.2 Backend Technology

The backend can be implemented using frameworks such as Node.js with Express or Python with Django. These frameworks provide scalable server-side architecture, API development capabilities, and security features.

6.3 Database

A relational database such as MySQL or PostgreSQL is suitable because the system requires structured relationships between entities such as alumni, degrees, employers, and organizations.

Relational databases also support strong data integrity, query optimization, and scalability.

6.4 Hosting

The system can be hosted on cloud platforms such as AWS, Azure, or Google Cloud. Cloud hosting allows scalability, reliability, and easy deployment of web applications.

7 Conclusion

The system design outlines the architecture, components, data model, and user interface structure of the Alumni Management System. The proposed design ensures scalability, maintainability, and secure data management while supporting the needs of administrators, viewers, and alumni users.